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POSTER

618A. ACUTE MYELOID LEUKEMIAS: BIOMARKERS AND MOLECULAR MARKERS IN DIAGNOSIS AND PROGNOSIS**ESAM is functionally involved in the activity of cancer stem-like cells in human acute myeloid leukemia**

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Abstract Despite major advances in molecularly targeted therapies for hematologic malignancies such as CML and malignant lymphoma, effective treatment options for acute myeloid leukemia (AML) remain limited. Drugs such as FLT3 inhibitors, DNA methyltransferase inhibitors, and BCL2 inhibitors have been introduced, yet relapse remains a major challenge. To improve outcomes, identifying novel molecular targets and developing corresponding therapies is crucial. Emerging evidence highlights the critical role of “leukemia stem cells” (LSCs), a small but resilient subpopulation, in disease relapse. Our previous studies and others have shown that AML LSCs are heterogeneous and dynamic (Stem Cells, 2021). This heterogeneity likely underpins their survival advantage under chemotherapeutic stress. A comprehensive understanding of LSC biology is therefore essential for developing targeted therapies.

In this study, we aimed to identify biomarkers that reflect the biological status of AML LSCs, and to explore strategies for modulating their activity. Using in-house screening methods, we identified multiple molecules implicated in hematopoietic regulation. One such molecule is endothelial cell-selective adhesion molecule (ESAM), a membrane protein primarily associated with the vascular endothelium. We previously reported that ESAM is expressed on normal hematopoietic stem cells (HSCs) (Blood 2009, Exp Hematol 2016) and serves as a marker of HSC activation in response to chemotherapeutic stress (J Immunol 2012, PLoS One 2016, J Exp Med 2021). Notably, ESAM is expressed in AML but not in lymphoblastic leukemia, and its expression differentiates proliferative from quiescent AML clones (Stem Cells 2022). These findings suggest that ESAM may serve as a functional biomarker of AML LSCs and a potential therapeutic target.

To investigate the role of ESAM in AML LSCs, we generated human AML cell lines expressing ESAM. The construct encoded ESAM fused at the C-terminus with AirID, a proximity biotinylation enzyme, enabling downstream interactome analysis. Among the transduced lines, we selected MOLM-14 cells, originally derived from relapsed AML with FLT3-ITD and lacking endogenous ESAM expression, for further study. Under normal culture conditions, ESAM expression did not alter proliferation compared to parental or AirID-only cells. However, upon treatment with FLT3 inhibitors (CEP701 and quizartinib), ESAM-expressing cells retained high drug sensitivity. Then, we performed RNA sequencing on MOLM-14 ESAM-AirID versus AirID-only cells. This analysis revealed marked transcriptomic changes: 680 genes were upregulated and 622 genes were downregulated (>3-fold change). Upregulated genes included PPP1R10, ZNF236, and P2RY6, all of which are associated with cell growth and stress responses. Downregulated genes included HSD17B8 and NTAN1, which are involved in steroid metabolism and protein degradation, respectively. Pathway analysis indicated that ESAM expression enhances signaling through G protein-coupled receptors and suppresses apoptosis-related pathways. These data support the hypothesis that ESAM promotes cellular activity of cancer stem-like cells in AML. Of interest, ESAM expression significantly upregulated MS4A3, recently identified as a regulator of LSC reactivation and drug sensitivity (Blood 2022). This further reinforces the functional relevance of ESAM in AML biology.

We next performed proximity-dependent biotinylation using the AirID system. Mass spectrometry identified 1,511 biotinylated proteins, including tight junction proteins ZO-1 and ZO-2 (encoded by TJP1 and TJP2), both of which contain PDZ domains that likely interact with the PDZ-binding motif of ESAM. Additionally, membrane-associated proteins such as ZDHHC17,

CNST, RASA3, and ATIC were detected in high abundance. These proteins are known to participate in diverse signaling and transport functions, suggesting that ESAM may influence multiple biological pathways through its intracellular interactions. In summary, our findings demonstrate that ESAM expression alters the transcriptomic and functional landscape of AML cells. ESAM enhances biological activity of AML LSCs, likely through modulation of signaling pathways and interactions with junctional and membrane-associated proteins. These findings identify ESAM as a key regulator of LSC behavior and a potential therapeutic target in AML.

Disclosure No relevant conflicts of interest to declare.

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